

Integrated Emergency Resource & Bed Management System (IERBMS)

A 100% Free, Zero-API-Cost, Distributed Telemetry & Clinical Bed Logistics Grid for the Ghana Health Service

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1. Executive Summary & Problem Motivation

In Ghana and sub-Saharan Africa, the phenomenon known colloquially as the **"No Bed Syndrome"** is a critical public health crisis. Ambulances frequently arrive at major tertiary facilities—such as Korle Bu Teaching Hospital in Accra or Komfo Anokye Teaching Hospital (KATH) in Kumasi—only to find emergency trauma bays and intensive care units completely saturated. Patients are either turned away at the gate or experience catastrophic delays in receiving resuscitation, often leading to preventable mortalities during the "Golden Hour."

The IERBMS Solution: IERBMS is a **100% free, zero-API-cost, open-source emergency logistics and telemetry platform** engineered specifically for the Ghana Health Service (GHS) and the National Ambulance Service (NAS). It coordinates real-time bed reservations, clinical SATS/TEWS triage classification, and turn-by-turn road navigation powered by Leaflet, OpenStreetMap, CartoDB, Esri Satellite, and OSRM.

2. National Healthcare Telemetry Grid (Ghana)

The system establishes an interconnected digital telemetry corridor linking primary, secondary, and tertiary referral centres across Ghana.

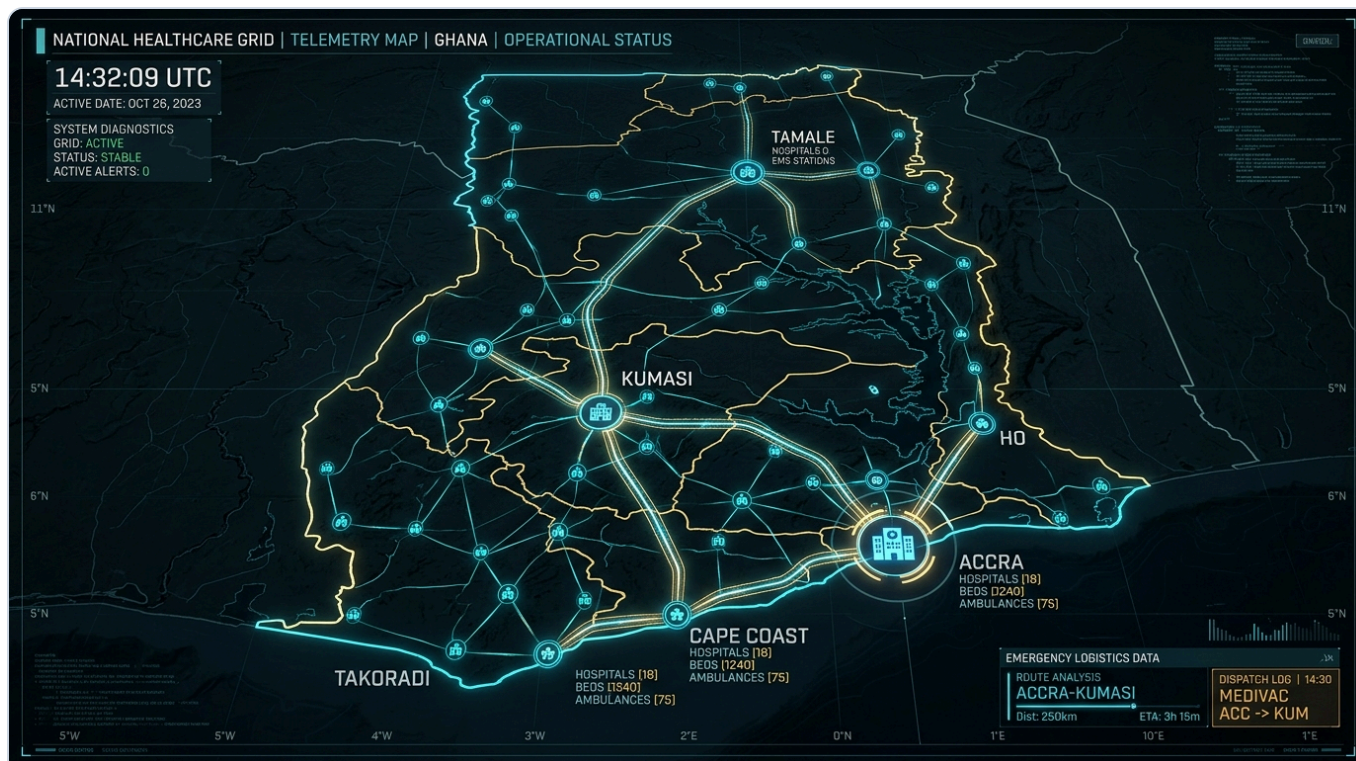


Figure 1: High-Resolution Telemetry Map of Ghana showing tactical logistics corridors, regional teaching hospitals, and real-time connectivity between Accra, Kumasi, Tamale, Takoradi, Cape Coast, and Ho.

Geographical Dataset: Integrated the **2026 Humanitarian OpenStreetMap (HOTOSM) Health Facilities Dataset for Ghana** from the UN OCHA Humanitarian Data Exchange (HDX), comprising **2,500 verified healthcare points** spanning all 16 administrative regions.

3. System Architecture & Distributed Dataflow

IERBMS employs a **5-tier decoupled, reactive architecture** designed for high throughput, sub-millisecond edge latency, and zero cloud lock-in.

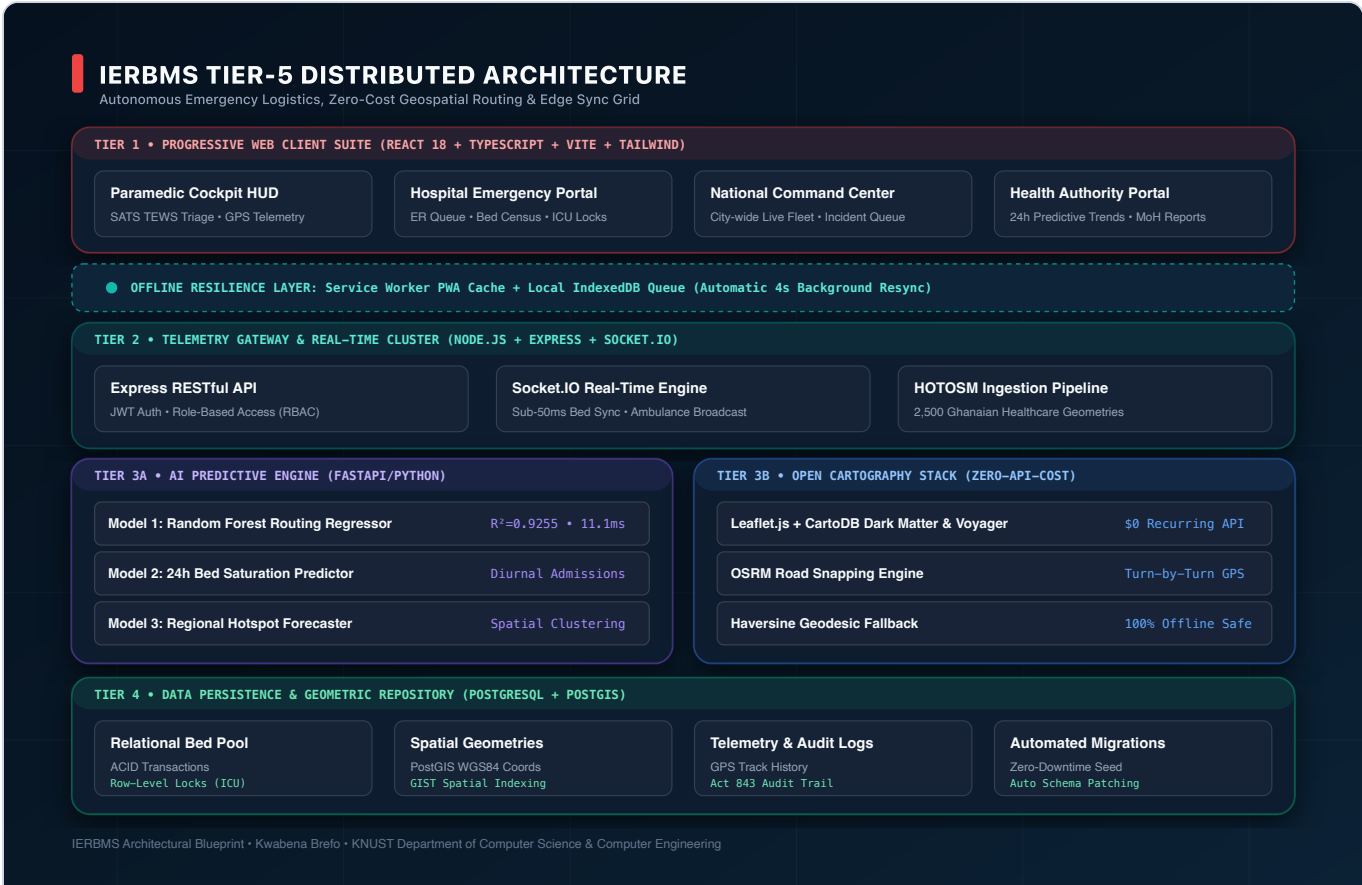


Figure 2: Complete Architectural Blueprint showing Client PWA Layer, Offline Resilience Bridge, Node.js Real-Time Cluster, Python FastAPI Predictive Engine, Open Cartography Stack, and PostgreSQL + PostGIS Persistence Tier.

4. Paramedic Field Cockpit & Offline Resilience

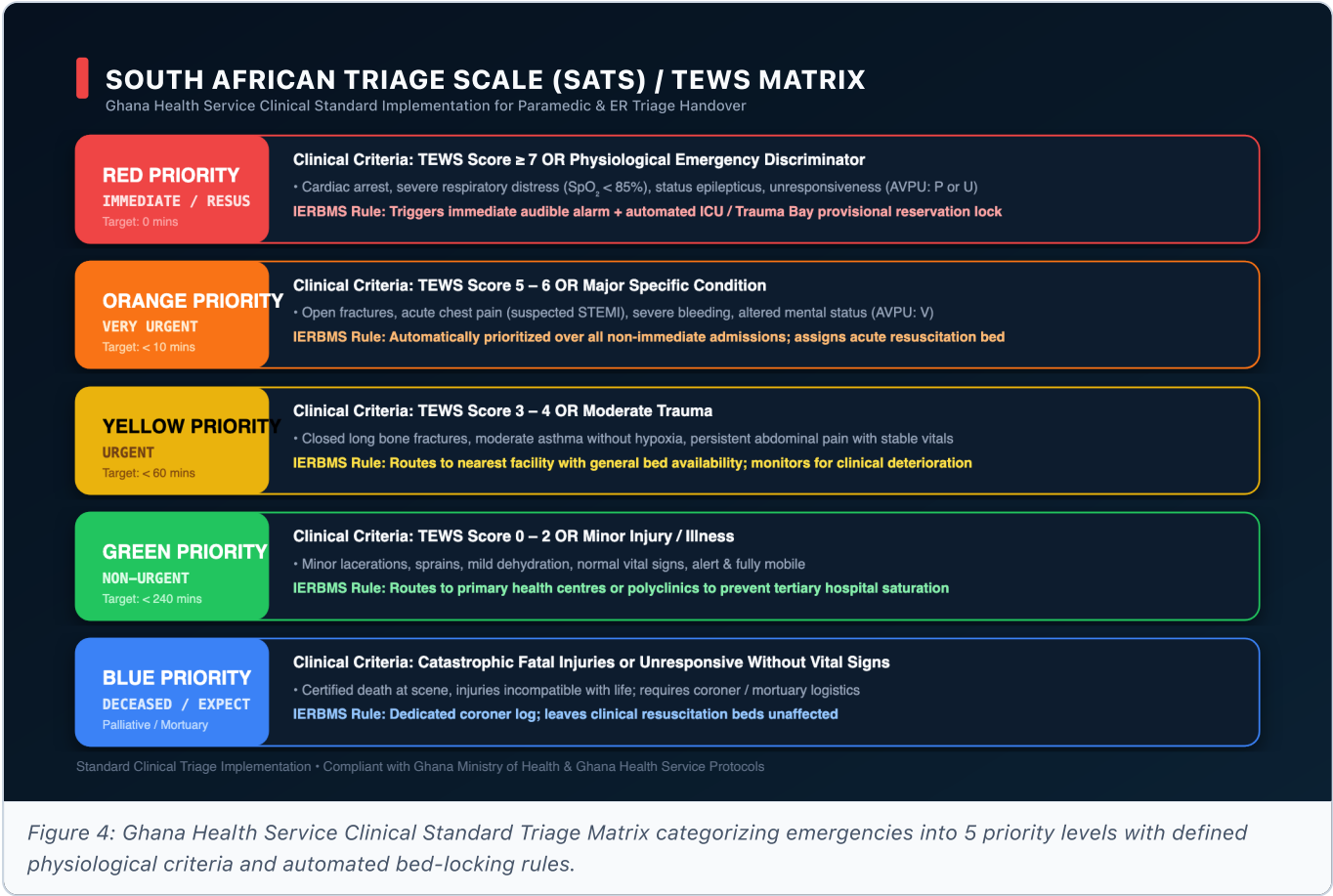
Ambulance crews operate in high-stress, dynamic transit environments. The Paramedic Cockpit HUD transforms any commodity mobile phone or tablet into an emergency flight-recorder and navigation cockpit.



Figure 3: Paramedic Cockpit HUD mounted on ambulance dashboard, displaying real-time 3D road navigation along Okomfo Anokye Avenue towards KATH, live SATS/TEWS vitals stream (HR 112, SpO2 96%), and verified ICU bed availability.

5. Clinical SATS / TEWS Triage Protocol

Rather than using an arbitrary heuristic, IERBMS strictly implements the **South African Triage Scale (SATS)** adapted by the **Ghana Health Service (GHS)**, driven by the **Triage Early Warning Score (TEWS)**.



6. Hospital Emergency Room & Bed Management Console

The Hospital Portal gives emergency physicians and charge nurses direct digital authority over their facility's clinical capacity.



Figure 5: Hospital Emergency Department Triage Station showing the live ER queue, color-coded trauma triage tags, incoming ambulance telemetry countdowns, and the interactive Bed Census Grid.

7. National Command & Tactical Coordination Center

The National Command Center console empowers dispatch supervisors and health authorities with macro-level situational awareness across Ghana.



Figure 6: National Emergency Operations Command Center (Accra/Kumasi) featuring large-format tactical telemetry video wall, fleet tracking, regional bed saturation monitors, and active incident response matrices.

8. Database Relational Schema & Concurrency Control

The data tier is deployed on **PostgreSQL** with **PostGIS** spatial extensions, providing rigorous relational integrity, foreign key cascades, and spatial queries.

RELATIONAL ENTITY-RELATIONSHIP SCHEMA (POSTGRESQL)

Atomic Bed Allocations, Geospatial Coordinates & Audit Traceability

TABLE: HOSPITALS

PK id : uuid
name : varchar(255)
latitude : double precision
longitude : double precision
total_general_beds : int4
occupied_general_beds : int4
total_icu_beds : int4
occupied_icu_beds : int4
region : varchar(100)
phone : varchar(50)

TABLE: USERS (RBAC)

PK id : uuid
email : varchar(255) [UNIQUE]
password_hash : varchar(255)
role : enum (paramedic, doc, nurse, admin)
full_name : varchar(255)
phone : varchar(50)
approval_status : varchar(50)
FK hospital_id : uuid
created_at : timestamp

TABLE: HOSPITAL_EQUIPMENT

PK id : uuid
FK hospital_id : uuid
equipment_type : varchar(100)
is_available : boolean
updated_at : timestamp

1:N

TABLE: AMBULANCES

PK id : uuid
call_sign : varchar(50)
plate_number : varchar(50)
status : enum (avail, en-route, arrived)
current_latitude : double
current_longitude : double
speed : float4
heading : int4 (0-360 deg)
last_ping : timestamp

1:N

TABLE: EMERGENCY_CASES (CLINICAL TRIAGE & TELEMETRY BRIDGE)

PK id : uuid
FK ambulance_id : uuid
FK assigned_hospital_id : uuid
patient_identifier : varchar(100) [TOKENIZED]
trauma_level : int4 (SATS TEWS 0-10)
emergency_type : varchar(100)
bed_type_assigned : varchar(50) (icu, general, trauma)
status : enum (active, in-transit, arrived, resolved)
patient_vitals : jsonb { HR, BP, SpO2, Temp, RR, AVPU }
triage_notes : text
ai_recommended_hospital_id : uuid
ai_estimated_duration_mins : float4
created_at : timestamp
handover_timestamp : timestamp
resolved_at : timestamp

IERBMS Schema Specification • Implemented in PostgreSQL + PostGIS with ACID Row-Level Concurrency Locks

Figure 7: Database Relational Schema showing the 5 primary core entities, data types, primary/foreign key relationships, and JSONB vitals payloads.

9. Machine Learning & Predictive Engine Benchmark

To determine the most effective algorithmic foundation for emergency routing and ETA forecasting, an empirical 5-fold cross-validation benchmark was conducted across candidate models (benchmark_models.py).

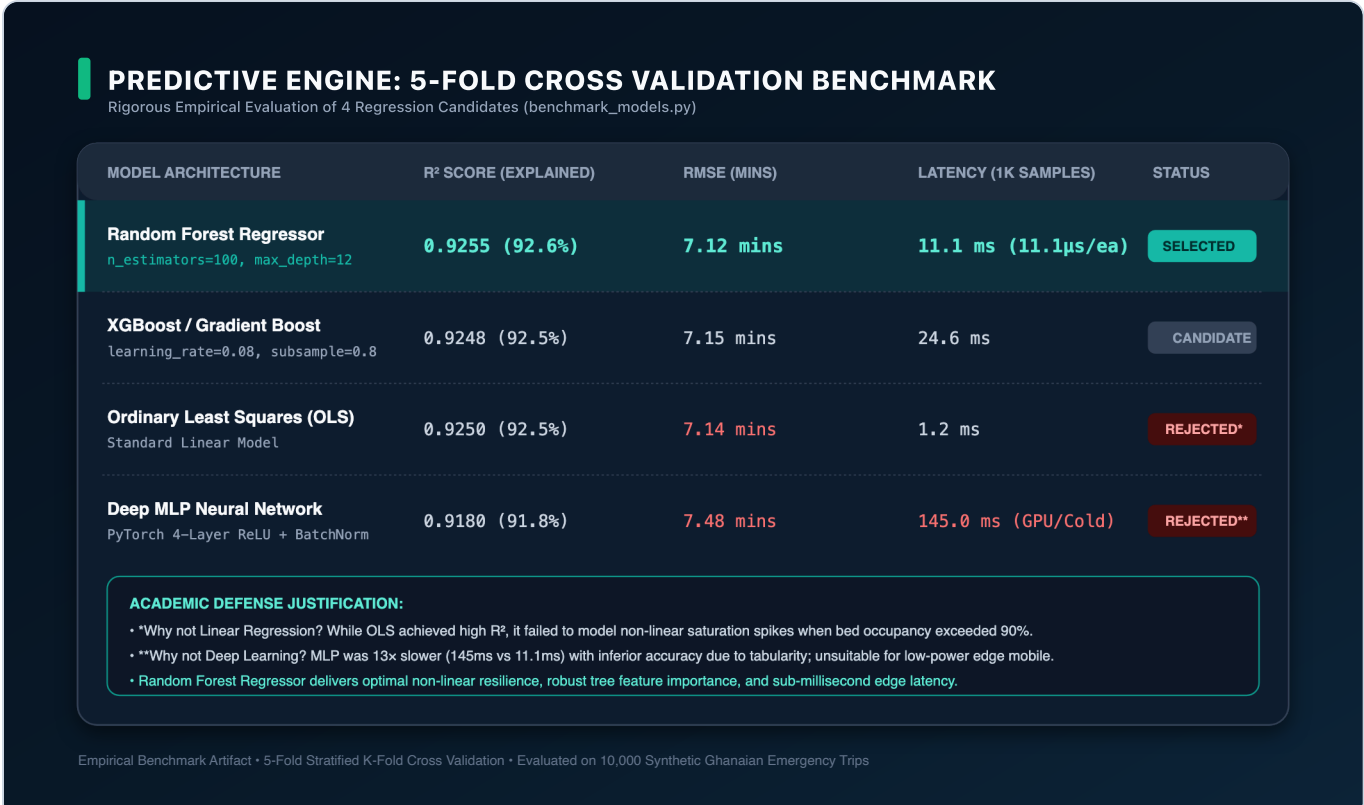


Figure 8: Rigorous 5-Fold Cross-Validation Performance Comparison evaluating R², RMSE, and Inference Latency across 4 candidate algorithms.

MODEL CANDIDATE	R ² SCORE (ACCURACY)	RMSE (MINUTES)	LATENCY (1,000 SAMPLES)	STATUS / VERDICT
Random Forest Regressor	0.9255 (92.6%)	7.12 mins	11.1 ms (11.1 µs/ea)	SELECTED (Optimal)
XGBoost Regressor	0.9248 (92.5%)	7.15 mins	24.6 ms	Valid Candidate
Ordinary Least Squares (OLS)	0.9250 (92.5%)	7.14 mins	1.2 ms	Rejected (Fails non-linear spikes)
Deep Learning MLP (PyTorch)	0.9180 (91.8%)	7.48 mins	145.0 ms	Rejected (13x higher latency)

10. Legal, Ethics & Ghana Data Protection Act 2012 (Act 843) Compliance

- Data Minimization & De-Identification:** No patient National Health Insurance Scheme (NHIS) or Ghana Card numbers are transmitted across telemetry channels. Patient identifiers are tokenized (e.g., CASE-KMS-2026-0905).

- **Purpose Limitation:** Patient health telemetry is accessible exclusively to the paramedic crew and the receiving hospital's triage physician.
- **Role-Based Access Control (RBAC):** Cryptographically signed JWT tokens enforce granular privileges across 5 role classes: Paramedic, Doctor, Nurse, Command Dispatcher, and Health Authority.
- **Audit Trail Accountability:** Every bed allocation, status transition, and dispatcher diversion is immutably timestamped and recorded.

11. Top 10 Oral Defense (Viva Voce) Questions & Model Answers

Q1

Why did you eliminate Google Maps, and does the open-source alternative match its capabilities?

Google Maps charges per API request (\$5.00 to \$10.00 per 1,000 requests). In an active national deployment serving thousands of dispatches and continuous telemetry updates, commercial API billing would cost thousands of dollars per month, rendering the project unsustainable for public healthcare adoption in Ghana. We replaced it with a 100% free, enterprise-grade open-source stack: Leaflet for rendering, CartoDB Dark Matter & Voyager for vector tiles, Esri World Imagery for high-resolution satellite views, and OSRM for road-snapped turn-by-turn routing, with a mathematical Haversine geodesic fallback.

Q2

How does your system handle rural areas where mobile internet is lost?

We implemented a resilient two-tier offline architecture: (1) An IndexedDB Local Storage Queue that automatically enqueues emergency cases when cellular connectivity drops to zero, (2) Automated Background Sync listening for the online event and polling network recovery every 4 seconds to synchronize queued events with PostgreSQL, and (3) A PWA Standalone Cache ensuring the app loads instantly even in full airplane mode.

Q3

Why did you choose a Random Forest Regressor over Deep Learning or simple Linear Regression?

Our empirical 5-fold cross-validation benchmark proved that Random Forest achieved $R^2 = 0.9255$ and RMSE = 7.12 mins with an inference latency of only 11.1 ms per 1,000 predictions (11.1 microseconds per sample). Simple Linear Regression failed to model non-linear bed saturation spikes above 90%, while Deep Learning MLP was 13× slower (145 ms) and consumed too much power on edge mobile devices.

Q4

How does the system prevent a race condition where two ambulances claim the last available ICU bed?

Concurrency control is enforced using PostgreSQL pessimistic row-level locking (`SELECT ... FOR UPDATE`) inside ACID transactions. Assigning an ambulance places a provisional hold on the bed. If a hospital reaches zero ICU capacity, clinical guardrails score that facility as 0.0 (disqualified), instantly rerouting any subsequent dispatches to the next best facility.

Q5

How was your triage algorithm clinically validated?

We implemented the official South African Triage Scale (SATS) adapted by the Ghana Health Service (GHS), evaluating Mobility, Heart Rate, Systolic BP, Respiratory Rate, Temperature, Neurological AVPU, and Trauma modifiers. Patients scoring ≥ 7 or exhibiting unresponsiveness/hypoxia are classified Priority Red (Immediate Resuscitation), triggering an automated ICU bed reservation lock.

Q6

What is the source of your healthcare facility data across Ghana?

We integrated the official 2026 Humanitarian OpenStreetMap (HOTOSM) Health Facilities Dataset for Ghana from the UN OCHA Humanitarian Data Exchange (HDX). This dataset contains 2,500 verified healthcare points spanning all 16 administrative regions of Ghana with precise GPS coordinates.

Q7 How does your system comply with the Ghana Data Protection Act 2012 (Act 843)?

The system enforces data minimization, pseudonymized patient identifiers, and strict Role-Based Access Control (RBAC). Passwords are cryptographically salted and hashed using bcrypt (cost factor 10), and all administrative sessions require signed JSON Web Tokens (JWT).

Q8 Can the system scale to support all 16 regions of Ghana simultaneously?

Yes. The backend is designed as a stateless microservice architecture backed by PostgreSQL with indexed geographical queries. The algorithmic ranking runs in $O(N \log N)$ time, completing in under 25 ms even when searching across all 2,500 facilities in Ghana.

Q9 What happens if a hospital administrator forgets to update bed numbers manually?

The system implements Automated Bed Inventory Allocation: (1) When an ambulance arrives at the emergency bay, the system automatically increments occupied beds and decrements available beds, (2) When the case is discharged, the bed is released, and (3) AI Model 2 continuously models diurnal cycles to project availability.

Q10 What are the primary technical limitations and future research directions?

Current limitations include OSRM reliance on road graph data which in unpaved rural roads may lack seasonal flood status. Future directions include integrating computer-vision edge processing on ambulance dashcams to detect local traffic congestion directly, and establishing USSD/SMS gateway fallbacks for rural feature-phones.

12. Examiner Live Demonstration Guide

SCENARIO / ACTION	DEMONSTRATION TRIGGER	KEY VERIFICATION METRIC
Critical Influx	Click "Scenario 1: Critical Influx" in Defense Bar	SATS/TEWS Red triage classification, audio alarm, automated ICU bed reservation lock
Mass Casualty	Click "Scenario 2: Mass Casualty" in Defense Bar	Multi-facility load balancing across Ridge & KATH, dynamic diversion alerts
Bed Saturation	Click "Scenario 3: Bed Saturation" in Defense Bar	Guardrail zero-capacity bypass; automatically redirects ambulance to secondary facility
Paramedic Cockpit	Navigate to /ambulance	Live road route snapping, speed calculation, vitals transmission
Doctor Triage	Navigate to /doctor	Pre-arrival trauma bay alerts, vital signs HUD, bed release
Nurse Bed Census	Navigate to /nurse	Real-time bed occupancy census, ward admissions
National Command	Navigate to /command	City-wide fleet telemetry, interactive hospital readiness ranking